Campus Laundry Booking System

Test Report

2 October 2026 · 171 automated tests · all passed

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1. Summary

The system was tested at four levels against the functional and non-functional requirements from the Phase 2 requirement analysis. **171 of 171 tests passed.** Testing also found four defects, which were fixed and retested (Section 5).

Test suite	What it tests	Tests	Result
Database integrity	Each rule is broken on purpose directly in MySQL; the database must refuse.	43	PASS 43/43
API / functional	Every feature through the real HTTP API as an admin and three students at once.	84	PASS 84/84
Browser / user interface	The real pages in Microsoft Edge: clicking, typing, dialogs, phone-sized screens.	34	PASS 34/34
Concurrency and performance	30 students at the same moment; response times.	9	PASS 9/9
Schema comparison	db/schema.sql builds exactly the same database as the migrations.	1	PASS (MATCH)
Total		171	ALL PASSED 171/171

> Every test creates its own temporary data (accounts on test-only email domains, a temporary laundry room) and deletes all of it afterwards, so the tests can be run safely on the real system at any time.

2. Test approach

2.1 Test levels

Level	Tool	How
Database integrity (unit level)	tools/test-constraints.js	Inserts test rows inside a transaction, then tries invalid statements straight against MySQL (bypassing the application) and checks the database refuses each one. Everything is rolled back.
API / functional (integration)	tools/test-api.js	Acts as an administrator and three students at the same time through the real HTTP API, covering every create, read, update and delete, permissions and workflows. Time-based steps (cycle finishing, missed slot, reminder) move the test's own bookings in time in the database, then check the scheduler reacts.
Browser / user interface (system)	tools/test-ui.js	Opens the real pages in Microsoft Edge (via Puppeteer) and clicks, types, answers dialogs and checks what appears on screen, as a student and an administrator in separate browser sessions, including a 375-pixel phone screen.
Concurrency and performance (non-functional)	tools/test-load.js	30 logged-in students send requests at exactly the same moment; response times are measured.

2.2 Test environment

Item	Version
Operating system	Windows_NT 10.0.26200 (x64)
Node.js	v24.19.0
MySQL	8.4.9
Browser	Microsoft Edge 154.0.4258.48
Application	Campus Laundry 1.0.0, http://localhost:3003

2.3 Pass criteria

- Functional tests: the expected result happens and invalid actions are refused with the right HTTP status and a clear message.
- Concurrency: exactly one of many simultaneous bookings of one slot succeeds, and nothing is lost when bookings do not conflict.
- Performance: 95% of availability-grid requests under 300 ms; 30 simultaneous "My laundry" requests answered within 2 seconds; pages load in under 200 ms.
- Usability on phones: no page scrolls sideways on a 375-pixel screen (only the booking grid scrolls inside its box).

3. Requirements coverage

Each requirement from the Phase 2 requirement analysis and the tests that cover it (a test can cover more than one requirement).

ID	Requirement	Tests	Result
FR1	User management: register and log in securely; Student and Admin roles	24	PASS
FR2	Booking: see available machines in real time; book a date and slot; prevent double bookings; cancel or reschedule	42	PASS
FR3	Notifications: reminder before the booking; laundry done; next user when a machine is free	29	PASS
FR4	Laundry tips: sorting, detergents, fabrics, maintenance	9	PASS
FR5	Feedback: rate the service or a machine; comments and issue reports; admin manages feedback	17	PASS
FR6	Administration: add, edit, remove machines; booking history and usage reports; act on feedback	45	PASS
NFR	Usability: simple, mobile-friendly	10	PASS
NFR	Performance: loads quickly; many users booking at once	9	PASS
NFR	Reliability: booking data and notifications accurate and consistent	44	PASS
NFR	Security: user data and passwords stored securely	23	PASS
NFR	Compatibility: web browsers and mobile devices	7	PASS

4. Results by suite

Database integrity (43/43)

`tools/test-constraints.js`

Area	Test	Result
UQ	UQ: duplicate email (uq_users_email)	PASS
UQ	UQ: duplicate student number (uq_users_student_number)	PASS
UQ	UQ: duplicate laundry room name (uq_laundry_rooms_name)	PASS
UQ	UQ: two machines with the same code in a room (uq_machines_room_code)	PASS
UQ	UQ: double booking of a machine slot (uq_bookings_machine_slot)	PASS
UQ	UQ: one student, two machines at the same time (uq_bookings_user_slot)	PASS
UQ	UQ: a cancelled booking frees its slot for someone else (slot_lock is NULL)	PASS
UQ	UQ: a machine can only have one open outage (uq_machine_outages_machine_open)	PASS
UQ	UQ: joining the same waiting list twice (uq_waitlist_entries_user_room_date_kind)	PASS
UQ	UQ: the same reminder twice for a booking (uq_notifications_booking_kind)	PASS
UQ	UQ: two tips with the same title (uq_laundry_tips_title)	PASS
FK	FK: booking a machine that does not exist (fk_bookings_machine)	PASS
FK	FK: feedback about a machine that does not exist (fk_feedback_machine)	PASS
FK RESTRICT	FK RESTRICT: delete a laundry room that has machines (fk_machines_room)	PASS
FK RESTRICT	FK RESTRICT: delete a machine that has bookings (fk_bookings_machine)	PASS
CHECK	CHECK: room closes before it opens (chk_laundry_rooms_hours)	PASS
CHECK	CHECK: 10-minute slots (chk_laundry_rooms_slot)	PASS
CHECK	CHECK: a 5-minute machine cycle (chk_machines_cycle)	PASS
CHECK	CHECK: a student without a student number (chk_users_student_has_number)	PASS
CHECK	CHECK: a student number with letters (chk_users_student_number)	PASS
CHECK	CHECK: invalid email (chk_users_email)	PASS
CHECK	CHECK: phone number with letters (chk_users_phone)	PASS
CHECK	CHECK: booking that ends before it starts (chk_bookings_times)	PASS
CHECK	CHECK: cycle running without a start time (chk_bookings_started)	PASS
CHECK	CHECK: start time without a finish time (chk_bookings_finishes)	PASS
CHECK	CHECK: finish time before the start (chk_bookings_finishes)	PASS
CHECK	CHECK: collected without a collection time (chk_bookings_collected)	PASS
CHECK	CHECK: cancelled without a cancellation time (chk_bookings_cancelled)	PASS
CHECK	CHECK: outage that ends before it starts (chk_machine_outages_period)	PASS
CHECK	CHECK: outage reason too short (chk_machine_outages_reason)	PASS
CHECK	CHECK: rating of 6 stars (chk_feedback_rating)	PASS
CHECK	CHECK: rating feedback without a rating (chk_feedback_rating_kind)	PASS
CHECK	CHECK: issue report without a description (chk_feedback_comment)	PASS
CHECK	CHECK: admin response without a response time (chk_feedback_response)	PASS
CHECK	CHECK: blank notification (chk_notifications_message)	PASS
CHECK	CHECK: tip text too short (chk_laundry_tips_body)	PASS
ENUM	ENUM: unknown booking status	PASS
ENUM	ENUM: unknown machine type	PASS

Area	Test	Result
ENUM	ENUM: unknown role	PASS
NOT NULL	NOT NULL: booking without a date	PASS
FK SET NULL	FK SET NULL: deleting a student keeps their booking history and feedback	PASS
FK CASCADE	FK CASCADE: deleting a student deletes their notifications and waiting-list places	PASS
FK CASCADE	FK CASCADE: deleting a machine or booking deletes its outages and notifications	PASS

API / functional (84/84)

`tools/test-api.js`

Area	Test	Result
Rooms	Create: admin adds laundry rooms	PASS
Rooms	Create: duplicate room name rejected	PASS
Rooms	Create: closing before opening rejected	PASS
Machines	Create: washers and dryers added, codes stored in capitals	PASS
Machines	Create: duplicate code in a room rejected	PASS
Machines	Create: cycle longer than the slot rejected	PASS
Machines	Update: cycle length changed	PASS
Accounts	Register: student number with letters rejected	PASS
Accounts	Register: student signs up with student number and home room	PASS
Accounts	Register: duplicate student number rejected	PASS
Accounts	Log in with the student number instead of the email	PASS
Admin	Students cannot manage machines	PASS
Booking	Read: availability grid has every machine × slot	PASS
Booking	Create: student books W1 tomorrow 10:00	PASS
Booking	Double booking of the same slot rejected	PASS
Booking	Same student, two machines at the same time rejected	PASS
Booking	A time that is not a slot start rejected	PASS
Booking	More than 7 days ahead rejected	PASS
Booking	A date in the past rejected	PASS
Booking	Read: other students see the slot as taken, without names	PASS
Booking	Read: the admin sees who booked	PASS
Booking	At most 3 open bookings per student	PASS
Reschedule	Update: booking moved to 12:00 and 10:00 is free again	PASS
Reschedule	Can't reschedule someone else's booking	PASS
Reschedule	The freed slot can be booked by another student	PASS
Waitlist	Join: second time rejected	PASS
Cancel	Cancel: student cancels a booking	PASS
Waitlist	Notify: the first student waiting for a dryer is told one is free	PASS
Waitlist	The waiting-list entry is marked as notified	PASS
Waitlist	Booking the machine removes the student from the waiting list	PASS
Cancel	A cancelled booking cannot be cancelled again	PASS
Out of order	Mark out of order: upcoming bookings on it cancelled	PASS
Out of order	Students get a "booking cancelled" notification with the reason	PASS

Area	Test	Result
Out of order	An out-of-order machine cannot be booked	PASS
Out of order	It shows as out of order in the grid	PASS
Out of order	Marking it twice rejected	PASS
Out of order	Back in service: it can be booked again	PASS
Out of order	The outage history is kept	PASS
Cycle	Can't start a booking for tomorrow yet	PASS
Cycle	Only the student who booked can start it	PASS
Cycle	Start: status in progress with a finish time 45 minutes later	PASS
Cycle	Starting twice rejected	PASS
Cycle	Collecting while it is still running rejected	PASS
Cycle	Done: the scheduler marks the finished cycle as done	PASS
Notifications	Done: the student is notified that the laundry is done	PASS
Cycle	Collected: status and collection time recorded	PASS
Notifications	Machine free: the next student booked on W1 today is told it is free	PASS
Booking	Read: history lists the collected and cancelled bookings	PASS
Notifications	Reminder: sent once, 30 minutes before the slot	PASS
No-show	Missed slot: marked as no-show after 15 minutes	PASS
No-show	The student is told the booking was released	PASS
No-show	The next student waiting for a washer that day is notified	PASS
Notifications	Mark one as read lowers the unread count	PASS
Notifications	Mark all as read	PASS
Notifications	Can't read someone else's notification	PASS
Notifications	Delete a notification	PASS
Tips	Read: students see the tips in all four categories	PASS
Tips	Drafts are hidden from students	PASS
Tips	Update: published tip visible to students	PASS
Tips	Students cannot write tips	PASS
Tips	Delete: tip removed	PASS
Feedback	A rating of 6 stars rejected	PASS
Feedback	An issue needs a room	PASS
Feedback	Create: issue linked to the machine and its room; students see only their own	PASS
Feedback	Admin sees all open feedback	PASS
Feedback	Dismissing needs an explanation	PASS
Feedback	Respond: status and reply saved, student notified	PASS
Feedback	Students cannot respond to feedback	PASS
Reports	Usage report: per machine, per room and per hour	PASS
Reports	Students cannot see the reports	PASS
Reports	Booking history filtered by room and student	PASS
Reports	Admin cancels a booking; the student is notified	PASS
Integrity	A room with machines cannot be deleted	PASS
Integrity	A machine with booking history cannot be deleted	PASS
Integrity	Slot length cannot change while there are upcoming bookings	PASS
Machines	Delete: a machine without bookings is deleted	PASS

Area	Test	Result
Machines	Retire: the machine leaves the grid and its upcoming bookings are cancelled	PASS
Admin	Deactivated student cannot log in	PASS
Admin	Deactivated student's session ends	PASS
Admin	Deactivating releases their upcoming bookings	PASS
Admin	Temporary password must be changed first	PASS
Admin	An admin without a student number cannot become a student	PASS
Admin	Admin can't demote themselves	PASS
Records	Deleting an account keeps its booking history	PASS

Browser / user interface (34/34)

`tools/test-ui.js`

Area	Test	Result
Admin	Log in as administrator → Dashboard	PASS
Machines	Add a laundry room with the form	PASS
Machines	Add a washer and a dryer with the form	PASS
Machines	Form validation: a cycle longer than the slot is refused with a message	PASS
Accounts	Register as a student with the form → My laundry	PASS
Accounts	Wrong password shows an error and stays on the login page	PASS
Accounts	Log in with the student number	PASS
Booking	Booking grid shows the room's machines and slots for tomorrow	PASS
Booking	Click a free slot → confirm bar → Confirm booking → My laundry shows it	PASS
Booking	The booked slot shows as "Yours" in the grid	PASS
Booking	Reschedule: pick a new slot → booking moved	PASS
Booking	Cancel: confirm the dialog → the booking moves to History as Cancelled	PASS
Waitlist	Join and leave the waiting list from the booking page	PASS
Tips	Tips page has the four categories; filtering by Fabric types	PASS
Feedback	Rate 4 stars with a comment → shown under My feedback	PASS
Feedback	Sending a rating without stars shows a message	PASS
Feedback	Report an issue about a machine	PASS
Profile	Update the phone number	PASS
Feedback	Admin replies to the issue and marks it resolved	PASS
Notifications	Student sees the reply: bell count and notification	PASS
Notifications	Mark all as read clears the bell	PASS
Machines	Mark a machine out of order (reason prompt) and back in service	PASS
Reports	Booking history: search by student	PASS
People	Find the student, deactivate and reactivate	PASS
Tips	Admin adds a tip with the form	PASS
Security	A student opening an admin page is sent back to their own page	PASS
Mobile	bookings.html fits a 375-pixel phone screen (no sideways scrolling)	PASS
Mobile	book.html fits a 375-pixel phone screen (no sideways scrolling)	PASS
Mobile	tips.html fits a 375-pixel phone screen (no sideways scrolling)	PASS
Mobile	feedback.html fits a 375-pixel phone screen (no sideways scrolling)	PASS

Area	Test	Result
Mobile	notifications.html fits a 375-pixel phone screen (no sideways scrolling)	PASS
Mobile	profile.html fits a 375-pixel phone screen (no sideways scrolling)	PASS
Mobile	The booking grid scrolls sideways inside its own box	PASS
Quality	No JavaScript errors on any page	PASS

Concurrency and performance (9/9)

`tools/test-load.js`

Area	Test	Result
Setup	30 students logged in at the same time	PASS
Concurrency	30 simultaneous bookings of one slot: exactly one succeeds 1 booked, 29 told "someone has just booked that slot", 1 row in the database	PASS
Concurrency	The losers get a clear message, not a server error	PASS
Concurrency	16 simultaneous bookings of different slots: all succeed and all are saved 16 rows	PASS
Concurrency	One student sending two bookings for the same time at once: only one is kept	PASS
Concurrency	5 simultaneous bookings by one student: the limit of 3 still holds	PASS
Performance	Availability grid: 95% of requests under 300 ms median 16 ms, 95th percentile 26 ms	PASS
Performance	30 students opening "My laundry" at the same moment: all answered within 2 s 132 ms in total	PASS
Performance	A page loads in under 200 ms 24 ms	PASS

5. Defects found during testing

These problems were found by the tests and the browser checks, fixed, and the tests run again.

ID	Found by	Symptom	Cause and fix	Retest
D-01	Concurrency test	When 16 students booked different slots at the same moment, only 3 bookings were saved; the others got a server error.	The booking transaction locked the student's existing bookings with SELECT … FOR UPDATE, which locks index ranges shared with other students, and MySQL detected deadlocks. Fixed by locking only the student's own users row, and by retrying a transaction automatically if MySQL ever reports a deadlock (lib/validate.js inTransaction).	PASS, 3 runs in a row
D-02	Concurrency test	Five simultaneous bookings by one student did not reliably apply the limit of three open bookings.	Same cause as D-01. With the student's row locked first, the bookings are counted one after the other.	PASS
D-03	Performance test	Occasionally 30 students opening "My laundry" at once took 3.4 seconds.	Opening MySQL connections costs up to a second when several open at once, and the connection pool closed idle connections after 60 seconds. Fixed by keeping idle connections for 10 minutes and opening them when the server starts (db/pool.js).	PASS (0.2–0.7 s), also after a 75-second idle pause
D-04	Browser check	When a booking was refused (e.g. over the limit), the error message disappeared immediately.	The page reloaded the grid after the error and the reload cleared the message. Fixed so the message is shown after the reload (public/book.html).	PASS (UI test "Booking")

6. How to run the tests

Start the app in one terminal, then run the tests in a second terminal in the project folder:

```
npm start

npm test
```

`npm test` runs every suite. They can also be run one at a time:

Command	Suite
npm run db:check	Schema comparison and database integrity (does not need the app running)
npm run test:api	API / functional
npm run test:ui	Browser / user interface (needs Microsoft Edge and `npm install` with dev tools)
npm run test:load	Concurrency and performance
npm run docs:test-report	Runs everything and rebuilds this report

1. Each suite prints PASS or FAIL for every test and a total at the end.
2. All test data is removed automatically, even when a test fails.